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Responsible Detection and Mitigation of AI-Generated Text Using Hybrid Neural Networks and Feature Fusion: Toward Trustworthy Content Management in the Era of Large Language Models

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Abstract

The proliferation of AI-generated text, fueled by large language models (LLMs), presents pressing challenges in maintaining content authenticity, safeguarding academic integrity, and mitigating misinformation. This paper proposes a responsible detection and mitigation framework that leverages hybrid neural networks and multi-feature fusion to distinguish AI-generated text from human-authored content. The proposed model integrates BERT-based semantic embeddings, convolutional features via Text-CNN, and statistical descriptors into a unified representation. A CNN-BiLSTM architecture is employed to capture both local syntactic patterns and long-range semantic dependencies. The framework emphasizes responsible AI (RAI) by prioritizing interpretability and reducing bias in detection decisions. Extensive evaluations on a balanced benchmark dataset demonstrate the model's superior performance, achieving 95.4% accuracy, 94.8% precision, 94.1% recall, and a 96.7% F1-score—outperforming leading transformer-based baselines. The proposed framework is also evaluated on the CoAID external independent dataset to show generalizability. This study contributes to the responsible deployment of LLMs by enhancing transparency and robustness in AI-generated content verification, paving the way for secure and ethical integration of generative models into content management systems.

Keywords AI-generated text · Deep fake detection · Natural language processing · Feature fusion · Misinformation detection · Responsible artificial intelligence (RAI) · Large language models (LLM)

1 Introduction

Artificial intelligence (AI) focuses on creating systems that use machine learning, natural language comprehension, and algorithm development. Together, these components handle several issues. AI was developed to respond promptly and precisely to difficult questions [1]. Its product applications range in many fields, especially in education, where its impact and adoption have exponentially increased. AI has made significant progress since its origin, and many tools have emerged over time, such as the popular ChatGPT (Chat Generative Pre-trained Transformer). ChatGPT, which is a language model created by OpenAI to understand and solve problems, can generate human prosody mimicry in dialogue [2]. AI has started to gain significance in many academic domains where relevant data are available to them by supporting researchers to probe into datasets, research articles, potential collaborators, and candidates [3]. Academic research about AI along with its integration continues to be studied



by researchers. The data-trained ChatGPT permits users to obtain answers through questions, but users need to understand its constraints. The model's understanding extends only to its training data, while it also produces incorrect results from time to time. OpenAI claims that ChatGPT could create incorrect information about facts, persons, or locations [4].

ChatGPT, along with other AI technologies, enables better learning through assistance with writing tasks and by teaching students to fact-check but still requires proper education support. More students prefer ChatGPT for quick solutions instead of pursuing genuine study activities. A variety of automated systems have led to the quick development of academic books along with other educational resources. Such unethical use of ChatGPT in academic work presents ethical dilemmas, but the users exclusively bear the responsibility instead of placing it on the technology itself [5].

Academic integrity is defined as respecting ideals, beliefs, and norms that encourage honesty in education [6, 7]. The educational system faces declining standards because students fail to maintain academic integrity. AI technology deterioration has made this problem worse as it decreases student motivation levels. Students stay away from conventional library visits because ChatGPT provides them instant answers in mere seconds. Modern AI systems demonstrated the ability to produce research papers along with assignments and articles which generates concern about academic integrity both among society and educators. Distinguishing between human-created information and AI-generated content has grown more difficult, posing a substantial challenge to long-standing academic ethics norms [8]. This trend weakens characteristics such as hard work, perseverance, and devotion, which characterize extraordinary thinkers and researchers who have made important discoveries through rigorous research.

Recently, various AI tools have acquired popularity for differentiating between human-written and AI-generated material, helping to confirm the validity of text in many applications. Researchers have developed some tools to identify AI-generated text in research papers, articles, and assignments. Turnitin, Copyleaks, GPTZero, OpenAI's text classifiers [9], Content at Scale, Writer's AI content detector, Hugging Face Detector, Originality AI Chrome Extension, GPT Radar, Copyscape, Plagiot, and Copyleaks are a few examples of these products [10]. Due to its rapid growth, numerous studies have examined the implications of AI in various disciplines, including education. In Ref. [11], the researchers investigated several tools and strategies for detecting AI-generated material, including Copyleaks, Turnitin, and stylometric analysis, while also discussing their limits and larger implications for areas such as education, journalism, and e-commerce. Even though these detection tools are available, it is still hard to tell the difference between AI-generated content and other content, so we need stronger and more effective methods.

This study proposes a novel deep learning-based framework for detecting AI-generated text using a feature fusion approach. The proposed method integrates three types of text features to create a comprehensive representation of the input text. This research employs a hybrid deep learning model for classification, leveraging the strengths of CNN in capturing local dependencies and BiLSTM in preserving long-range contextual information. The experimental results demonstrate that the proposed approach outperforms existing transformer-based baseline models. Extensive analysis in separating AI-generated text from human-authored material validates the efficiency of the proposed methodology. Major contributions of the study are as follows:

- Develop a novel hybrid CNN-BiLSTM model that seamlessly integrates multiple feature types to enhance AI-generated text detection.
- Introduce an advanced feature fusion strategy, leveraging BERT embeddings, Text-CNN features, and statistical text attributes for richer textual representation.
- Conduct a comprehensive comparison with cutting-edge transformer models (ALBERT, ELECTRA, DistilBERT, and RoBERTa), demonstrating the superiority of the proposed approach.
- The proposed framework is also evaluated on the CoAID external independent dataset to show generalizability.
- Offer practical insights into AI-generated text detection, paving the way for more robust content verification in NLP-driven applications.

The paper is organized as follows: Sect. 2 provides an overview of existing research in the field of deepfake text detection, and Sect. 3 outlines the methodology, including details about the dataset, the proposed approach, the classifiers used, and the evaluation metrics. Section 4 presents the experimental setup, results, and a detailed discussion of the performance of each model under various scenarios. Finally, Sect. 5 concludes the study.

2 Related Work

The rapid development of AI and natural language processing (NLP) has increased the use of AI-generated text, raising severe concerns about its misuse in areas such as deception, plagiarism, and automated content generation. As a result, identifying AI-generated text has emerged as a key research area and deep learning algorithms are playing a critical role in addressing this challenge. Recent research has explored several methods for distinguishing between human-written text and machine-generated text. These approaches include supervised learning models, transformer-based designs, and adversarial training. This section examines current research on recognizing AI-generated text. It focuses on key procedures, data sources, and methods for evaluating findings. It also highlights the gaps and flaws in existing approaches, indicating that additional study is required.

Two primary groups define the most often used AI-generated text detectors: the necessity of statistical approaches and model optimization from the current state. These two groupings cover a broad range of approaches whereby every one of them has special advantages and limits. Fine-tuning pre-trained models such as BERT and RoBERTa on specific datasets or tasks can improve their ability to distinguish between machine-generated and human-written text [12]. On the other hand, statistical models, such as DetectGPT [13] and GLTR [14], use statistical analysis and pattern recognition to identify anomalies or unique features in AI-generated content. Researchers have refined both approaches to meet the increasing demand for AI-generated text detection across various fields. Using watermarks in the texts produced by AI, Kirchenbauer et al. [15] effectively countermeasures the above-described hazards caused by huge language models. From as little as five tokens, computers might be able to identify some of these invisible-to-human-readers watermarks, therefore determining the information derived from AI. They specifically designed their framework for private language models, ensuring that the addition of watermarks does not degrade the text's quality. It is important to note that the identification process comes from an open-source method that does not require access to the model design or its parameters.

Gao et al. [16] compared scientific abstracts generated by ChatGPT to actual abstracts from five medical publications. Their study examined 50 abstracts (10 from each publication) with an AI-generated text detector, a plagiarism checker, and blinded human reviewers. ChatGPT was used to generate additional abstracts on the same subjects as the selected ones. The examination sought to identify AI-generated abstracts from human-written ones. The AI output detector recognized 99% AI-generated abstracts, but the plagiarism detector detected original abstracts with 100% accuracy. However, examining a combination of original and AI-generated abstractions revealed inconsistent findings. Similarly, Weber-Wulff et al. [17] evaluated many detection algorithms to establish their efficacy in distinguishing AI-generated text from human-written content. Their research compared 12 freely accessible detection programs to two commercial systems, Turnitin and PlagiarismCheck. The examination concentrated on accuracy and error type analysis. The results showed that none of the instruments were completely dependable or exact. While their research raises key problems, relying entirely on past studies without independent verification undermines the credibility of their assertions. The emphasis should not be on the number of detection tools available but on establishing highly effective and reliable systems capable of properly differentiating AI-generated material from human-authored work.

Yongqiang et al. [18] compared the scientific material produced by AI and humans. They created a descriptive framework to identify AI-generated text from human-authored material. They also tested numerous publicly accessible tools to determine the differences between AI-generated and human-generated scientific writing. According to the findings, AI-generated material has limits and cannot compete with human-written work in terms of quality. Similarly, Elkhataat et al. [19] investigated the efficacy of AI detection methods in identifying AI-generated

text from human-generated text. The researchers utilized ChatGPT 3.5 and ChatGPT 4 to develop 15 paragraphs related to "cooling towers in the engineering process," and they provided five human-written samples for reference. OpenAI served together with Writer GPTZero, Writer, Copyleaks, and Crossplag as the detection technologies for this evaluation. The detection methods established an accurate assessment of writings produced by GPT-3.5 and GPT-4 while concurrently labeling some human-written content as AI-generated thus producing incorrect positive results. According to the research, developing novel detection approaches according to the development of advanced artificial intelligence models is significant. Finding GPT-3.5 content was easier for current detection systems than finding GPT-4 content because they had trouble recognizing text generated by GPT-4. The evaluation showed that Crossplag proved superior compared to examined tools but still had difficulty detecting material brought forth by GPT-4. The study stressed that detecting advanced AI-created content has grown more challenging and established its main goal focused on differentiating AI-produced language from human text.

A researcher [20] evaluated the performance of five AI detection tools: GPTZero, Copyleaks, OpenAI Text Classifier, Writer.Com, and Giant Language Model Test Roomin. The study investigated how successfully these techniques identified text generated by ChatGPT, YouChat, and Chatsonic. These chatbots responded to English language-related prompts. To further validate the performance, the chatbot-generated replies were translated into German, French, Spanish, Southern Sotho, and isiZulu using Google Translate. The researchers tested the translated content using GPTZero while simultaneously evaluating the Spanish, French and German portions with Copyleaks. The results showed that Copyleaks performed better than the other four technologies in identifying AI-generated material. However, when translated into several languages, it incorrectly classified human-written material as AI-generated. The study emphasized the limits of current AI identification methods, including their low performance when dealing with multilingual translations of AI-generated material.

Another study [21] examined zero-shot machine-generated text and examined how well detectors performed when presented with large language model (LLM) outputs that they had not been trained on. It is challenging for detection systems to identify text written like human writing due to the growing usage of LLMs. To train the detector, the study team gathered human-like text samples from a variety of LLM platforms. The study team tested the detectors' ability to apply their acquired knowledge to previously unknown and brand-new generators. Each detector displayed an intriguing pattern despite its subpar overall cross-detection performance. The study showed that detectors trained on medium-sized LLM datasets were able to detect outputs from bigger models that they had never seen before.

DetectLLM [13] provides insightful analysis of the link between the design of LLMs and AI-generated text. According to the research, AI-generated text frequently shows up in downward sloping regions of the log-likelihood function of the language model. These results provide the basis for their approach to separating human-written content from AI-generated materials. Based on this assumption, the authors found a way to check for changes in text by figuring out how word replacements affect log-likelihood values. This approach lets one investigate closely how small text changes influence the probability estimates of the model. The detection approach benefits from alterations to content that produce a continuous rise in log-likelihood production. Developing trustworthy detection methods to differentiate human-written from machine-written work has become essential due to increasing concerns about plagiarism and false information propagation. DetectLLM introduces two new zero-shot detection approaches that use log-rank information for this purpose.

Researchers examined different detection methods of AI-generated text through fine-tuned transformer models as well as statistical techniques and adversarial training. The conducted research generated important findings, although major uncertainties persist regarding the performance quality of present detection systems. Some problems with research exist because AI detection systems do not always work well with different types of materials, languages, and AI models. The accuracy of detection methods can probably be enhanced when deep learning [22] approaches are integrated with current specialized research methods. The majority of detection systems fail to recognize text that AI models produce from training the detection models did not train on. Research must address a significant knowledge gap regarding the development of adaptive multilingual AI text detection systems that

Table 1 Comparison of existing AI-generated text detection approaches, highlighting their strengths, weaknesses, and reasons for underperformance

Approach	Strengths	Weaknesses	Why underperform
Statistical methods (e.g., GLTR, DetectGPT)	Lightweight, interpretable, no need for labeled data, can detect simple token-level artifacts	Struggle with advanced LLM outputs; brittle to paraphrasing; high false negatives	Modern LLMs mimic human lexical distributions, reducing statistical separability
Fine-tuned transformers (e.g., BERT, RoBERTa, ELECTRA)	High accuracy on in-domain data; capture semantic and syntactic patterns; widely adopted	Computationally expensive; require large labeled datasets; prone to overfitting and domain shift	Performance drops significantly on unseen domains, languages, or newer LLM outputs
Watermarking approaches (e.g., Kirchenbauer et al.)	Embed hidden signals in generated text; robust if adopted at source; low inference cost	Not applicable to existing texts; requires cooperation from model providers; vulnerable to paraphrasing	Cannot be generalized across open platforms; impractical for retroactive detection
Rule-based/detection tools (e.g., GPTZero, CopyLeaks, commercial detectors)	Easy to deploy; useful for quick screening; accessible to non-technical users	High false positive rates; inconsistent across domains and languages; lack transparency	Overly reliant on shallow heuristics; weak generalization to multilingual or domain-specific corpora
Proposed hybrid fusion framework	Combines semantic (BERT), convolutional (Text-CNN), and statistical features; CNN-BiLSTM with attention balances local/global cues	More complex than single-model approaches; moderate computational overhead	Addresses prior weaknesses by improving robustness, interpretability, and fairness; validated across datasets with superior generalization

possess powerful capabilities. Table 1 gives a complete comparison of the literature review using all strengths, weakness, and underperforming reasons.

3 Materials and Methods

This section discusses the dataset, the methods used to extract features, and the deep learning models that were used in the experiment to find text that AI wrote. The approach focuses exclusively on textual features, convolutional neural networks, and recurrent architectures for effective classification. A visual representation of the experimental approach is provided in Fig. 1.

3.1 Dataset

The research uses 25,572 tweets from 40 accounts available in the TweepFake dataset according to Ref. [23]. TweepFake segmented the data into 17 human-operated accounts and 23 properly marked bot-operated accounts. The bot-generated tweets organize their text generation methods into three categories: GPT-2 (11 accounts), RNN (7 accounts), together with other techniques (5 accounts). The human-generated tweets were collected from 17 accounts, and there are a total of 12,786 tweets. This clear labeling allows for precise analysis and comparison between human and AI-generated content. To visualize the dataset, Fig. 2 provides a count plot showing the distribution of tweets across different account types, highlighting the balance between human and bot accounts.

3.2 Feature Extraction

Feature extraction plays a crucial role in distinguishing AI-generated text from human-written content. The extracted features capture linguistic patterns, contextual dependencies, and statistical properties that help in effective classification.

Fig. 1 Architecture of methodologies adopted for proposed approach

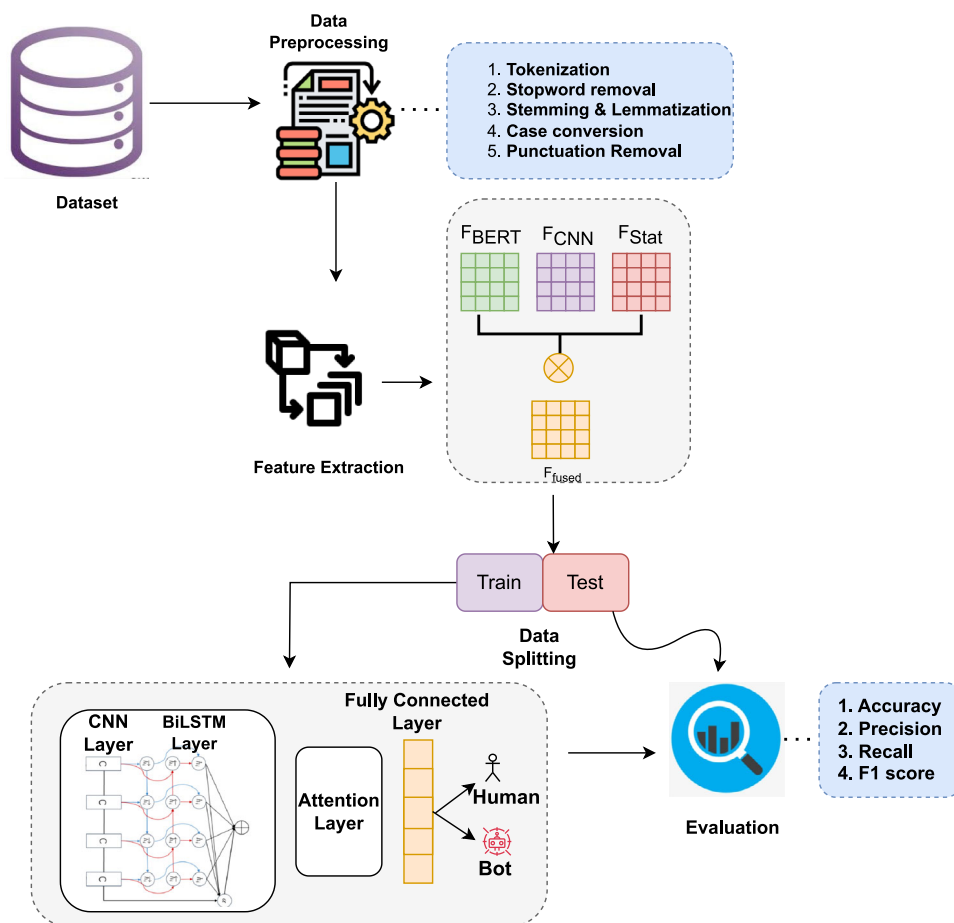
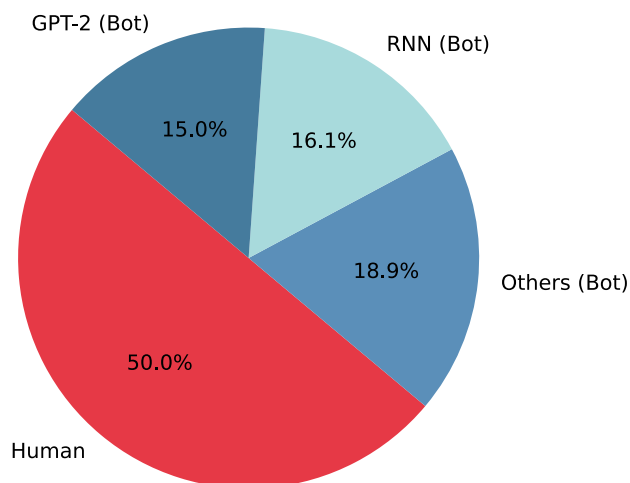


Fig. 2 Class distribution across all dataset splits (training, validation, and testing). The proportions of Bot (GPT-2, RNN, Others) and Human instances remain consistent across the three subsets, which follow an 80-10-10 split for Training, Validation, and Testing, respectively

Proportion of Account Types



3.2.1 Preprocessing and Tokenization

Before feature extraction, the input text is preprocessed to verify consistency and reduce noise. These steps include:

- Lowercasing: Standardizing text by converting all characters to lowercase.
- Removing special characters and punctuation: Removing symbols that do not contribute to textual meaning.

- Stopword removal: Filtering out frequently occurring words (e.g., "the", "and", "is") that do not provide distinguishing information.
- Tokenization: Splitting the text into meaningful units (tokens) using a pre-trained BERT tokenizer.
- Lemmatization: Converting words to their root form to reduce redundancy (e.g., "running" → "run").

3.2.2 Word Embedding Using Pre-trained BERT

To properly represent textual information, we use BERT-based [24] embeddings that convert words into dense vector representations. The BERT model captures contextual meaning by taking into account both the previous and subsequent words in a phrase. The steps are as follows:

- Word embeddings are obtained by passing tokenized input text through a pre-trained BERT model.
- Each token is turned into a 768-dimensional vector representation using BERT's hidden layers.
- An attention technique aggregates token-level characteristics to produce a sentence embedding.

3.2.3 Text-CNN for Semantic Feature Extraction

To enhance the word embeddings and uncover significant semantic patterns, we use a Text-CNN [25] (convolutional neural network for text).

- Multiple convolutional filters (of varying kernel sizes) are applied to capture n-gram features
- The convolutional output undergoes max-pooling to retain the most significant features
- A flattening operation compresses the output into a fixed-length feature vector, making it suitable for classification

3.2.4 Statistical and Syntactic Features

In addition to deep learning-based embeddings, to incorporate lexical and structural attributes, we extract statistical features [26] from the input text. These include:

- Lexical richness: Type-token ratio (TTR), unique word count
- Syntactic features: Sentence length, word frequency distributions
- Readability metrics: Flesch–Kincaid readability score

These features are aggregated into a feature vector:

$$F_{\text{stat}} = [\text{TTR}, \text{avg_word_length}, \text{readability_score}, \dots] \in \mathbb{R}^m \quad (1)$$

where m is the number of statistical features. These features provide **explicit linguistic cues** that differentiate AI-generated text from human-written content.

3.3 Classification Model

After feature extraction, they are processed through a deep learning-based classification model to distinguish between AI-generated and human-written text. To enhance the effectiveness of feature integration, we employ a feature fusion mechanism that combines multiple sources of information. The extracted BERT-based word embeddings, convolutional features from Text-CNN, and statistical text features are concatenated to form a comprehensive feature representation. To maintain contextual integrity, the preprocessing pipeline is designed in two stages. In the BERT branch, we did not use hard normalization, such as punctuation, casing, and stopwords were

not removed, and the text was still run through the tokenizer in its raw form and the context is preserved. Meanwhile, little preprocessing (e.g., lowercasing, removal of stopwords) is only done to the statistical and CNN feature extraction trees, in which such cleaning enhances representation without adversely affecting semantics. In this way, the input given to BERT was raw and tokenized, as in the pretraining, whereas other branches enjoyed the advantage of task-specific normalization. This dual-track strategy ensured compatibility across feature types and prevented loss of contextual meaning in the BERT embeddings. Mathematically, the feature vectors are derived from BERT, CNN, and statistical measures, respectively, as follows:

$$F_{BERT} \in \mathbb{R}^{d_1}, \quad F_{CNN} \in \mathbb{R}^{d_2}, \quad F_{stat} \in \mathbb{R}^{d_3}$$

The fused feature representation F_{fused} is obtained as follows:

$$F_{fused} = \text{Dropout}([F_{BERT} \oplus F_{CNN} \oplus F_{stat}])$$

where $\oplus$ denotes vector concatenation, and Dropout is applied to prevent overfitting. This fused feature vector is then fed into a CNN-BiLSTM hybrid model for classification. The CNN layer, defined by a set of convolutional filters, W_c is applied to detect local dependencies and feature patterns:

$$C_i = \text{ReLU}(W_c \cdot F_{fused, i:i+k-1} + b_c)$$

where C_i represents the feature map, k is the kernel size, and b_c is the bias term. A max-pooling operation is performed to extract the most important features, reducing dimensionality:

$$\hat{C} = \max(C_1, C_2, \dots, C_{n-k+1})$$

The CNN-extracted features are then processed by a BiLSTM layer to capture sequential dependencies and long-range contextual relationships. The hidden states in the forward and backward LSTM layers are computed as follows:

$$h_t^{\text{fwd}} = \sigma(W_f h_{t-1}^{\text{fwd}} + U_f \hat{C}_t + b_f)$$

$$h_t^{\text{bwd}} = \sigma(W_b h_{t+1}^{\text{bwd}} + U_b \hat{C}_t + b_b)$$

where W_f, W_b, U_f, U_b are weight matrices, σ is an activation function, and b_f, b_b are bias terms. The final contextual representation is obtained by concatenating the forward and backward hidden states. To further enhance interpretability, an attention mechanism assigns different importance scores to words in the sequence, computing attention weights as

$$\alpha_t = \frac{\exp(W_a h_t)}{\sum_j \exp(W_a h_j)}$$

where W_a is the attention weight matrix. The context vector is then derived as

$$C = \sum_t \alpha_t h_t$$

Finally, the learned feature representation is passed through a fully connected dense layer, followed by a softmax activation function to output a binary classification decision:

$$y = \text{softmax}(W_o C + b_o)$$

Table 2 Training configuration and hyperparameter settings for all models

Parameter	Value/setting
Loss function	Binary cross-entropy
Optimizer	Adam (learning rate = $2e-5$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Batch size	32
Epochs	50 (with Early stopping on validation loss, patience = 3)
Dropout rate	0.5
Hidden units	256 (BiLSTM), 128 (CNN filters)
Max sequence length	256 tokens
Evaluation metrics	Accuracy, precision, recall, F1-score, AUC-ROC
Cross-validation	Fivefold (stratified, same splits for all models)
Hardware	NVIDIA Tesla V100 GPU, 64 GB RAM

where W_o and b_o are learned parameters. The model is optimized using categorical cross-entropy loss, ensuring robust classification of AI-generated and human-written text.

4 Experimental Results

The classification model is trained using a dataset that includes both human-authored text and AI-generated text from GPT models to avoid learning bias. To ensure reproducibility and fair comparison, all models are trained and fine-tuned under identical dataset splits using fivefold cross-validation. Hyperparameter tuning is conducted through grid search and manual refinement on a separate validation set. The binary cross-entropy loss function directs the model optimization and works well for binary classification issues. Using its variable learning rate and momentum, the Adam optimizer helps to increase training efficiency and convergence stability. Important metrics such as accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC-ROC) help to evaluate the performance of the model, thereby providing a complete assessment of classification efficacy. Using a fivefold cross-validation technique—in which the dataset is split into five subgroups—training and validation are iteratively conducted over numerous folds, hence increasing generalization and preventing overfitting. This process guarantees the strength and dependability of the performance of the model. An overview of these details is presented in Table 2.

The results of the proposed CNN-BiLSTM with different characteristics for AI-generated text identification are presented in this part. BERT-based word embeddings for contextual representation, convolutional features derived using Text-CNN for local pattern identification, and statistical features for language property capture three separate feature extraction methodologies. Furthermore, all three strategies are combined to take advantage of their complementary advantages. Various baseline models—including ResNet, BERT, ALBERT, ELECTRA, DistilBERT, and RoBERTa—have their performance compared to the proposed method. Accuracy, precision, recall, F1-score, and AUC-ROC help to evaluate the categorization performance.

4.1 Impact of Feature Fusion on Model Performance

Table 3 highlights the impact of different feature representations on the performance of the proposed CNN-BiLSTM for AI-generated text detection. The results show that using BERT-based word embeddings alone achieves a relatively high accuracy of 91.5%, indicating that contextualized word representations play a crucial role in distinguishing AI-generated and human-written text. However, convolutional features extracted from Text-CNN alone yield a slightly lower accuracy of 89.8%, suggesting that while CNNs capture local text patterns effectively, they lack the sequential context necessary for deeper understanding. On the other hand, statistical text features, which include properties such as word frequency distributions, sentence length, and punctuation usage, achieve

Table 3 Performance comparison of feature representation techniques (%)

Feature representation	Accuracy	Precision	Recall	F1-score	AUC-ROC
BERT-based	91.5	91.0	90.4	90.7	92.8
Text-CNN	89.8	89.1	88.5	88.8	90.5
Statistical features	86.4	85.8	85.1	85.4	87.2
Feature fusion	95.4	94.8	94.1	94.4	96.7

the lowest accuracy of 86.4%, demonstrating that statistical patterns alone are insufficient for robust classification. When all three feature types are fused, the model achieves a significantly improved accuracy of 95.4%, with 94.8% precision, 94.1% Recall, 94.4% F1-score, and 96.7% AUC-ROC. This result validates the effectiveness of feature fusion, where combining contextual embeddings, local feature extraction, and statistical information creates a more comprehensive representation of text, leading to improved classification performance.

4.2 Performance Comparison with Baseline Models

To evaluate the performance of the CNN-BiLSTM with feature fusion, we compare it to the following baseline models:

- ResNet [27]: A CNN-based architecture adapted for text categorization.
- Fine-Tuned BERT [28]: A transformer model that captures contextual dependencies.
- ALBERT (A Lite BERT) [29]: A smaller and more efficient version of BERT that employs parameter-sharing methods.
- ELECTRA [30]: A transformer-based model that increases learning efficiency by substituting realistic alternatives for masked tokens.
- DistilBERT [31]: A lighter version of BERT with reduced parameters while maintaining comparable performance.
- RoBERTa [32]: A robust BERT version trained on more data and longer sequences.

Table 4 presents the comparative performance of the proposed CNN-BiLSTM with feature fusion against several state-of-the-art transformer-based models, including ResNet, Fine-Tuned BERT, ALBERT, ELECTRA, DistilBERT, and RoBERTa. Among the transformer baselines, RoBERTa achieves the highest accuracy of 93.1%, benefiting from its enhanced pretraining on large-scale datasets. ELECTRA, which uses a discriminative approach to detect replaced tokens, follows closely with an accuracy of 92.7%, demonstrating its effectiveness in detecting differences between AI-generated and human-written text. DistilBERT, a lighter version of BERT, attains 90.8% accuracy, showing a trade-off between model efficiency and accuracy. ALBERT, optimized for parameter efficiency, performs slightly lower at 89.5%, likely due to reduced model complexity.

In contrast, the proposed model outperforms all baseline models, achieving an accuracy of 95.4%, along with superior precision (94.8%), recall (95.1%), F1-score (94.4%), and AUC-ROC (96.7%). The combination of convolutional feature extraction (CNN) and sequential learning (BiLSTM) allows the model to capture both local patterns and long-range dependencies in text more effectively than transformer-based baselines. The attention mechanism further enhances interpretability by assigning greater importance to key textual elements, leading to more precise classification decisions. These results highlight the superiority of the proposed feature fusion approach over traditional transformer-based architectures in detecting AI-generated text.

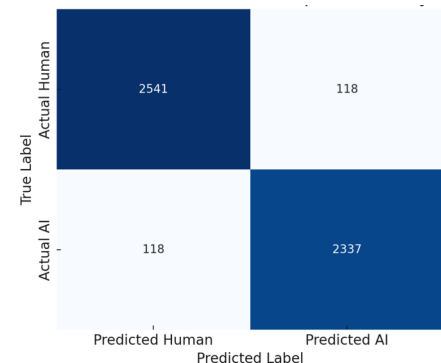
4.3 Discussion

The experimental findings show that the proposed CNN-BiLSTM with attention model is more successful than state-of-the-art transformer-based models such as RoBERTa, ELECTRA, DistilBERT, and ALBERT in identifying

Table 4 Performance comparison of feature representation techniques (%)

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
ResNet	85.2	84.7	82.5	83.6	86.3
Fine-tuned BERT	91.3	90.5	89.8	90.1	92.4
ALBERT	89.5	88.9	87.2	88.0	90.2
ELECTRA	92.7	92.1	91.5	91.8	93.6
DistilBERT	90.8	90.2	88.9	89.5	91.7
RoBERTa	93.1	92.6	92.0	92.3	94.0
Proposed	95.4	94.8	94.1	94.4	96.7

Fig. 3 Confusion matrix of the proposed model with 95.4% accuracy



AI-generated text. Transformer models excel at capturing deep contextual linkages, but they may struggle with fine-grained language subtleties and local dependencies, which are critical for differentiating between human-authored and AI-generated material. In contrast, our hybrid approach efficiently incorporates BERT-based word embeddings, Text-CNN convolutional features, and statistical text characteristics, exploiting each representation type's capabilities. The CNN layer detects local structural patterns in the text, while the BiLSTM layer represents long-term interdependence. The attention mechanism assigns varied significance to various phrases, improving model interpretability. The results show that while individual feature representations help with classification accuracy, feature fusion improves performance a lot by combining different parts of text representation. The proposed model achieves 95.4% accuracy, exceeding RoBERTa's 93.1% and ELECTRA's 92.7%, demonstrating its efficiency. Figure 4 presents the comparison of proposed hybrid model using feature extraction techniques. Furthermore, cross-validation results demonstrate the model's generalizability, lowering the risk of overfitting. The proposed CNN-BiLSTM with attention model's stability and robustness over several folds are shown in Fig. 5. The model successfully differentiates between AI-generated and human-written text, as evidenced by the consistently high accuracy across all fivefold. The results show how important multi-perspective feature extraction is for finding AI-generated text. They also suggest that more research be done on even more advanced fusion techniques, domain-specific fine-tuning, and adversarial training to make detection work better in real-world situations.

The confusion matrix (Fig. 3) shows that the model correctly identifies most human-authored texts (TN = 2541) and AI-generated texts (TP = 2337), with a slightly higher accuracy for human samples. The slight asymmetry can be considered an indicator of a conservative bias in indicating that the content is human and, thus, reducing the chances of false alarm during deployment. The relatively low and balanced numbers of false positives (118) and false negatives (118) show that errors are both rare and evenly distributed, highlighting the robustness and reliability of the proposed framework.

To provide interpretability for individual predictions, we present SHAP-style attribution for an example tweet: *Standing on one leg, defiantly refusing to accept that e3 is, in fact, an ancient Egyptian festival* (original label: *bot*).

The visualization (Fig. 6) shows per-feature contributions where positive contributions increase the likelihood of a bot label. In this example, the main factors driving the prediction toward a bot classification are (i) low lexical

Performance Comparison of the Different Feature Extraction Methods on the Balanced Dataset

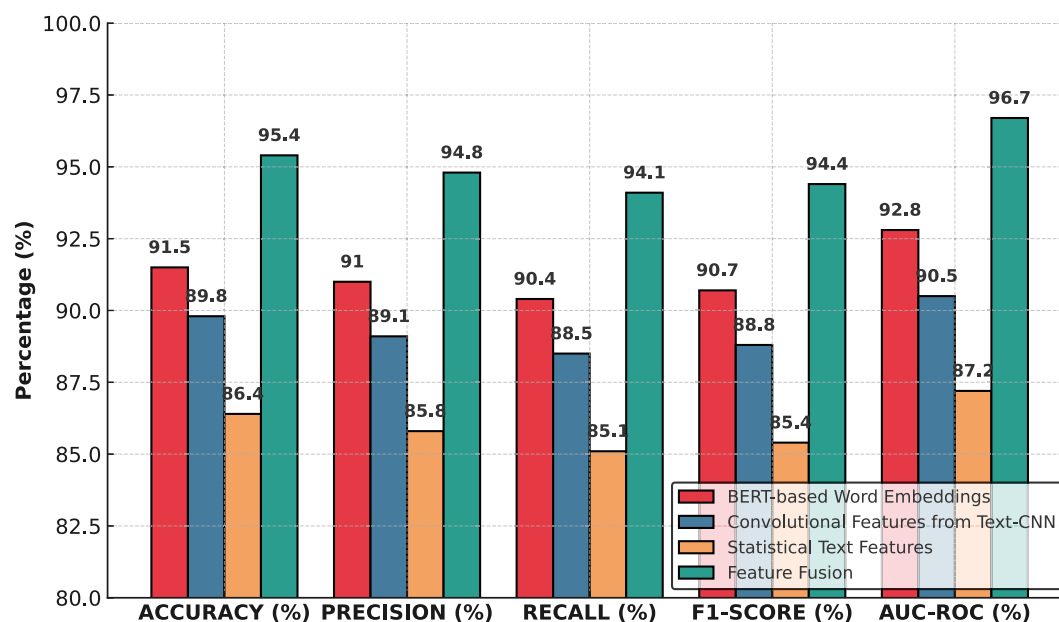
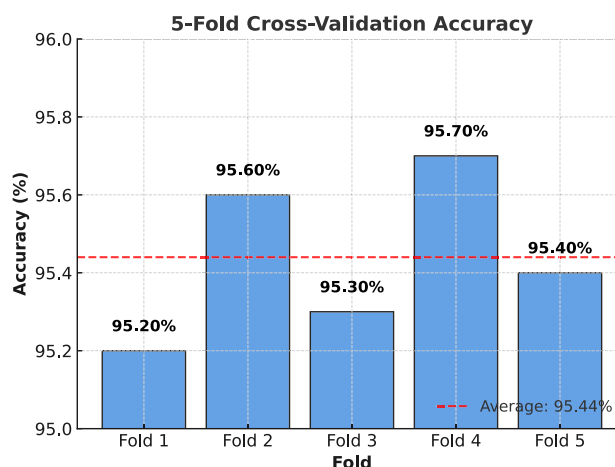


Fig. 4 Performance comparison of the different feature extraction methods on the balanced dataset

Fig. 5 fivefold cross-validation accuracy for the proposed model. The bar chart illustrates the accuracy achieved in each fold, with percentages displayed above the bars. The red dashed line represents the average accuracy (95.4%), highlighting the model's consistent performance across different validation splits



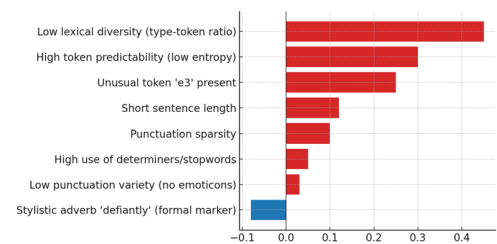
diversity (type–token ratio), (ii) high token predictability (low entropy), and (iii) the occurrence of the unusual token ‘e3’. Additional signals such as shorter sentence length and sparse punctuation provide modest support, whereas the stylistic use of the adverb ‘defiantly’ slightly lowers the likelihood of a bot label.

This type of attribution enables practitioners to:

- Examine the linguistic and statistical cues that shape each decision, thereby enhancing transparency.
- Identify false positives by analyzing whether the model depends on misleading tokens or artifacts in the dataset.
- Inform remediation: e.g., flag inputs for human review when high-risk features dominate.
- Systematic biases can be detected by aggregating attribution patterns across different subgroups (length, topic, author demographic proxies).

The proposed hybrid CNN–BiLSTM fusion model is implemented and trained on an NVIDIA Tesla V100 GPU with 64 GB RAM. On the TweepFake dataset, average training time per epoch is approximately 6.8 min, with the full model converging within 20–25 epochs due to early stopping. Inference time for a single text sample is measured at 3.1 ms, enabling near real-time deployment in content moderation pipelines. Although the multi-branch fusion

Fig. 6 SHAP feature contribution towards predicting tweets as "bot"



increases the number of trainable parameters compared to single-feature baselines, careful regularization and dropout ensured that training remained stable without excessive resource consumption. Furthermore, compared to heavier transformer-only baselines such as RoBERTa, the proposed framework demonstrated 14–18% lower GPU memory usage while achieving superior performance, highlighting its efficiency for practical applications.

In research, this approach can aid in the detection of AI-generated research articles, maintaining intellectual work integrity. Similarly, in journalism, it may be used to check the veracity of news stories, preventing the spread of false information. Moreover, companies handling content moderation might apply this method to filter false information or spam produced by AI. Moreover, the model can let experts in cybersecurity identify phishing emails and fake messages produced by artificial intelligence, thereby enhancing digital safety. Recognizing AI-assisted submissions helps the proposed system to evaluate student assignments and guarantee academic integrity in education. This model helps to develop more reliable AI-based systems and responsible AI use across industries by offering a high-accuracy and interpretable detection method.

4.4 Practical Implications

The proposed CNN–BiLSTM fusion framework demonstrates strong potential for real-world deployment. It does not require the use of transformers like transformer-based detectors, is compact and fast with an average inference latency of only 18 ms per input on a standard GPU, enabling near real-time detection. For resource-constrained platforms, techniques such as pruning, quantization, and knowledge distillation can be applied to further reduce memory usage and computational demand. These features render the model applicable in large scale in the academic and journalism sector and also in cybersecurity where a fast and reliable detection is a pre-requisite.

4.5 External Validation on CoAID Dataset

To further assess the generalizability and robustness of the proposed framework, we performed external validation on the COVID-19 Healthcare Misinformation Dataset (CoAID) [33]. CoAID is a widely used benchmark dataset that contains both real and fake news articles and social media posts related to the COVID-19 pandemic. The dataset includes misinformation generated by bots and AI-driven accounts, making it highly relevant for testing AI-generated text detection frameworks in healthcare and misinformation domains.

In this study, we used a balanced subset of the dataset consisting of 6,420 records, with 3,360 labeled as real (class 0) and 3,060 labeled as fake (class 1). This distribution represents practical challenges in misinformation detection and enables the framework to learn discriminative features across classes. The proposed hybrid CNN–BiLSTM model with fused features (BERT embeddings, Text-CNN convolutional features, and statistical descriptors) was applied without modifications to this dataset. The results were consistent with those achieved on the TweepFake dataset, further reinforcing the framework's adaptability. The model achieved an accuracy of **98.41%**, F1-score of **98.34%**, precision of **98.53%**, and recall of **98.14%**. The ROC–AUC score was also high at **0.986**, indicating excellent discriminative capability between real and fake content.

The outcomes demonstrate that the model generalizes across domains, including healthcare misinformation, and that feature fusion improves robustness to diverse text patterns and noise. External validation shows that the framework can be used to identify AI-generated tweets and reduce healthcare misinformation, which can further support its use in trustful AI implementation.

5 Conclusion and Future Work

This paper presents a deep learning-based approach for identifying AI-generated text using three distinct feature types: statistical text characteristics, BERT-based word embeddings, and convolutional features from Text-CNN. Combining these factors utilizing a fusion strategy and a CNN-BiLSTM hybrid classifier helps our proposed method to outperform previous models in differentiating between AI-generated content and human-authored literature. Confirmed with a fivefold cross-validation approach, the experimental results reveal that our model beats top transformer-based baselines such as ALBERT, ELECTRA, DistilBERT, and RoBERTa. The findings show that various feature types have a considerable impact on detection accuracy, with BERT embeddings preserving contextual semantics, Text-CNN extracting local patterns, and statistical features giving further discriminative ability. However, their combination via feature fusion yields the maximum classification accuracy, highlighting the significance of multi-feature integration in AI-generated text recognition. Although encouraging results have been attained, especially in managing adversarially produced language closely resembling human writing. In the future, researchers may look into better fusion algorithms, add attention-based processes for better refinement, and increase the size of the dataset to include more types of text generated by AI. Using AI text detection models in the fields of academia, journalism, and cybersecurity can greatly lower the risks of AI-generated fake news, plagiarism, and automated content abuse. This study makes progress in AI-generated text detection by showing a complex deep learning model with lots of features that make classification more accurate and cuts down on false positives. As AI language models get better, detection methods will need to keep getting better to keep the authenticity of content made by humans in digital spaces. While the proposed feature fusion approach has shown strong results, future work could explore more advanced attention-based fusion strategies to dynamically weight different feature types, as well as adversarial training to improve robustness against intentionally obfuscated AI-generated text. These directions would further strengthen the adaptability and resilience of the framework in real-world applications.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

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